

Table 3: Results of several methods on AVA benchmark measured in terms of binary classification accuracy of aesthetic labels.

Method	Backbone	Labels in the pre-training	Aesthetic label alone	Results (%)
DMA-Net (?)	AlexNet	$\sim 10M$	×	75.4
AA-Net (?)	VGG		×	76.9
(?)	AlexNet		×	77.3
MNA-CNN (?)	VGG		×	77.4
NIMA (?)	Inception		✓	81.5
Pool-3FC (?)	Inception		✓	81.7
A-Lamp (?)	VGG		×	82.5
MP_{ada} (?)	ResNet-18		✓	83.0
Our pretext task	AlexNet	0	✓	82.0
+ non-linear layers	ResNet-18			82.8

Table 4: The degradation caused by discarding some image manipulation operations during the pretext task training.

Attribute	AVA	AADB
Much noise	$\sim 0.1\%$	$\sim 0.3\%$
Poor lighting	$\sim 0.3\%$	$\sim 0.4\%$
Soft / Grainy	$\sim 0.4\%$	$0.5 \sim 0.8\%$
Fuzzy	$\sim 0.1\%$	$\sim 0.2\%$
Camera shake	$0.3 \sim 0.5\%$	$0.4 \sim 0.6\%$
Distracting	$\sim 0.2\%$	$0.3 \sim 0.4\%$

dynamics, while the two-stage learning scheme is more stable and consequently leads to better results.

Image editing operations. We randomly select one operation in Tab 1 and exclude it from the pre-training stage to analyze the impact on the final assessment performance. We can make several observations from the corresponding results shown in Tab. 4. First, editing operations which are related to softness, camera shake, and poor lighting are relatively more significant in learning aesthetic-aware features, compared to other operations. Second, image manipulations which are related to distracting, fuzziness, and noise have relatively smaller margins. This fact indicates that these operations may create aesthetically ambiguous instances that do not always provide consistent signals. Interestingly, camera shake has the most significant impact on the AVA, while soft/grainy are the most important ones for the AADB.

Entropy-based weighting. Without our entropy-based weighting scheme, there will be at least 0.5% down in the performance after task adaptation. Besides, some undesired dynamics will occur during the training process. Therefore, we can strengthen meaningful training signals from manipulated instances and improve training efficiency by assigning an entropy-based weight to each patch.

Conclusions

In this paper, we propose a novel self-supervised learning scheme to investigate the possibility of learning useful aesthetic-aware features without manual annotations. Based

on the correlation between negative aesthetic effects and several expert-designed image manipulations, we argue that an aesthetic-aware representation space should distinguish between results yielded by various operations. To this end, we propose two pretext tasks, one to recognize what kind of editing operation has been applied to an image patch, and the other to capture fine-grained aesthetic variations due to different manipulation parameters. Our experimental results on three benchmarks demonstrate that the proposed scheme can learn aesthetics-aware features effectively and generally outperforms existing self-supervised counterparts. Besides, we achieve results comparable to state-of-the-art supervised methods on the AVA dataset without using labels from ImageNet. We hope these findings can help us obtain a better understanding of image visual aesthetics and inspire future research in related areas.

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